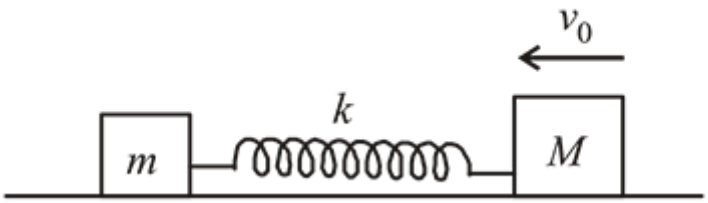
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two blocks of masses m and M are joined with an ideal spring of spring constant k and kept on rough surface as shown. The spring is initially unstretched and the coefficient of friction between the blocks and the horizontal surface is μ. What should be the maximum speed of the block of mass M such that the smaller block does not move?



(A) $\mu g\sqrt{\frac{Mm}{(M+m)k}}$

(B) $\mu g\sqrt{\frac{(M+m)k}{Mm}}$

(C) $\mu g\sqrt{\frac{(2M+m)m}{kM}}$

(D) None of these

Q.2

A ball of mass m moving with a constant velocity strikes against a ball of same mass at rest. If e = coefficient of restitution, then what will be the ratio of velocity of two balls after collision?

(A) $\frac{1-\mathrm{e}}{1+\mathrm{e}}$

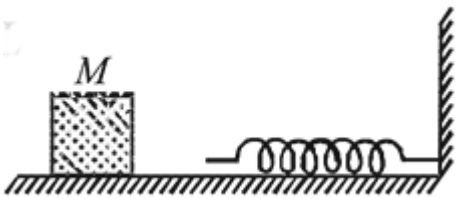
(B) $\frac{\mathrm{e}-1}{\mathrm{e}+1}$

(C) $\frac{1+\mathrm{e}}{1-\mathrm{e}}$

(D) $\frac{2+\mathrm{e}}{\mathrm{e}-1}$

Q.3

The block of mass M moving on the frictionless horizontal surface collides with the spring of spring constant k and compresses it by length L. The maximum momentum of the block after collision is



(A) $\frac{kL^2}{2M}$

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(B) $\sqrt{MkL}$

(C) $\frac{ML^2}{k}$

(D) zero

Q.4

An object of mass $2.0\ \text{kg}$ makes an elastic collision with another object of mass M at rest and continues to move in the original direction but with one-fourth of its original speed. What is the value of M ?

(A) $0.75\ \text{kg}$

(B) $1.0\ \text{kg}$

(C) $1.2\ \text{kg}$

(D) None of these

Q.5

A bullet of mass $20\ \text{g}$ and moving with $600\ \text{m/s}$ collides with a block of mass $4\ \text{kg}$ hanging with the string. What is velocity of bullet when it comes out of block, if block rises to height $0.2\ \text{m}$ after collision ?

(A) $200\ \text{m/s}$

(B) $150\ \text{m/s}$

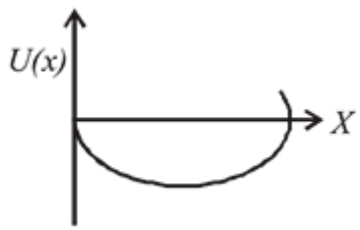
(C) $400\ \text{m/s}$

(D) $300\ \text{m/s}$

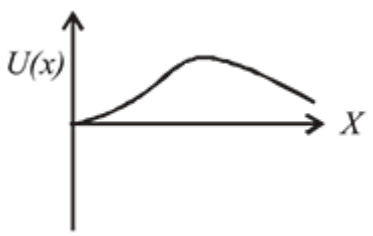
Q.6

A particle, which is constrained to move along the x -axis, is subjected to a force in the same direction which varies with the distance x of the particle from the origin as $F(x) = -kx + ax^3$. Here k and a are positive constants. For $x \geq 0$, the functional form of the potential energy $U(x)$ of the particle is

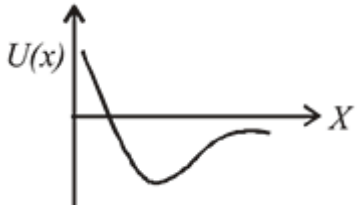
(A)



(B)

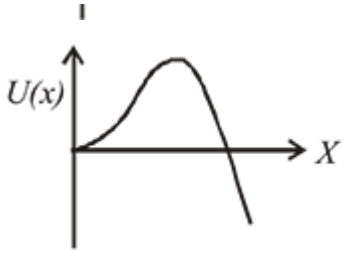


(C)



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(D)



Q.7

A block of mass $0.50\ \text{kg}$ is moving with a speed of $2.00\ \text{ms}^{-1}$ on a smooth surface. It strikes another mass of $1.00\ \text{kg}$ and then they move together as a single body. The energy loss during the collision is

(A) $0.16\ \text{J}$

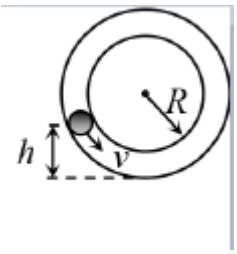
(B) $1.00\ \text{J}$

(C) $0.67\ \text{J}$

(D) $0.34\ \text{J}$

Q.8

With what minimum speed v must a small ball should be pushed inside a smooth vertical tube from a height h so that it may reach the top of the tube? Radius of the tube is R. (Assume radius of cross-section of tube is negligible in comparison to R)



(A) $\sqrt{2g(h+2R)}$

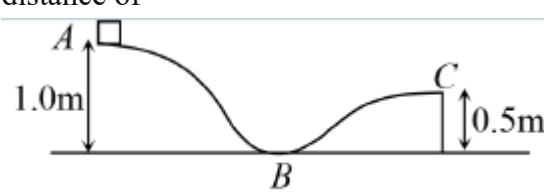
(B) $\frac{5}{2}R$

(C) $\sqrt{g(5R-2h)}$

(D) $\sqrt{2g(2R-h)}$

Q.9

A particle slides on a frictionless elevated curved track from a point A which teminates in a straight horizontal section ending at C as shown. If the particle looses the contact with the track at C, the particle will hit the ground from C at a horizontal distance of



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(A) $0.5\ \mathrm{m}$

(B) $0.8\ \mathrm{m}$

(C) $1.0\ \mathrm{m}$

(D) $1.2\ \mathrm{m}$

Q.10

A rod of mass m and length ℓ is made to stand at an angle of 60° with the vertical. Potential energy of the rod in this position is

(A) $\mathrm{mg}\ell$

(B) $\frac{\mathrm{mg}\ell}{2}$

(C) $\frac{\mathrm{mg}\ell}{3}$

(D) $\frac{\mathrm{mg}\ell}{4}$

Q.11

A 15 gm ball is shot from a spring gun whose spring has a force constant of $600\ \mathrm{N/m}$. The spring is compressed by $5\ \mathrm{cm}$. The greatest possible horizontal range of the ball for this compression is $\left(g = 10\ \mathrm{m/s^2}\right)$

(A) $6.0\ \mathrm{m}$

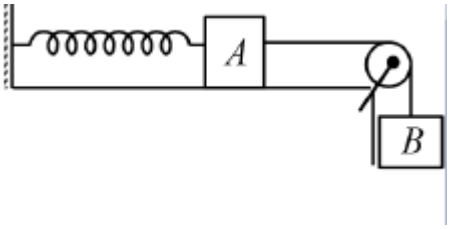
(B) $12.0\ \mathrm{m}$

(C) $10.0\ \mathrm{m}$

(D) $8.0\ \mathrm{m}$

Q.12

In the adjoining figure block A is of mass m and block B is of mass $2\ \mathrm{m}$. The spring has a force constant k. All the surfaces are smooth and the system is released from rest with spring unstretched, then



(A) The maximum extension of the spring is $\frac{4mg}{k}$

(B) The speed of block A when extension in spring is $\frac{2mg}{k}$, is $2g\sqrt{\frac{m}{k}}$

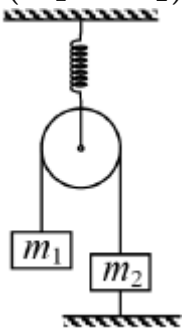
(C) Net acceleration of block B when the extension in the spring is maximum, is $\frac{g}{2}$.

(D) Tension in the thread for extension of $\frac{2mg}{k}$ in spring is mg.

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Q.13

In the figure shown, pulley and spring are ideal. If k is spring constant of spring, the potential energy stored in it is $(m_1 > m_2)$



(A) $\frac{2m_1^2g^2}{k}$

(B) $\frac{2m_2^2g^2}{k}$

(C) $\frac{(m_1+m_2)^2g^2}{k}$

(D) $\frac{1}{2}\frac{(m_1-m_2)^2g^2}{k}$

Q.14

When a body is projected vertically up from the ground with certain velocity, its potential energy and kinetic energy at a point A are in the ratio 2 :
3. If the same body is projected with double the previous velocity, then at the same point A the ratio of its potential energy to kinetic energy is

(A) $9:1$

(B) $2:9$

(C) $1:9$

(D) $9:2$

Q.15

A straight bar, of mass $15\ \mathrm{kg}$ and length $2\ \mathrm{m}$, at rest on a frictionless horizontal surface, receives an instantaneous impulse of 7.5 Ns perpendicular to the bar. If the impulse is applied at the mid point of the bar, the energy transfered is



(A) $3.2\ \mathrm{J}$

(B) $1.9\ \mathrm{J}$

(C) $3.8\ \mathrm{J}$

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(D) $2.5\ \mathrm{J}$

Q.16

A particle is released from the top of two inclined rough surfaces of height h each. The angle of inclination of the two planes are 30° and 60° respectively. All other factors (e.g. coefficient of friction, mass of block etc.) are same in both cases. Let K_1 and K_2 be the kinetic energies of the particle at the bottom of the plane in two cases. Then $\left(K_1 \text{ for } \theta = 30^\circ \text{ and } K_2 \text{ for } \theta = 60^\circ, \mu < 1/\sqrt{3}\right)$

(A) $K_1 = K_2$

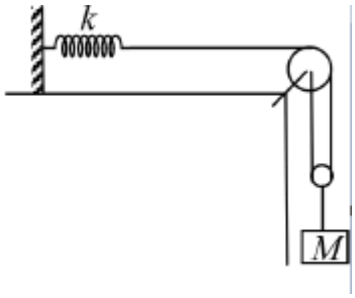
(B) $K_1 > K_2$

(C) $K_1 < K_2$

(D) data insufficient

Q.17

A block of mass m is released from rest when the k extension in the spring is x_0. The maxinum downward displacement of the block is



(A) $\frac{Mg}{2k} - x_0$

(B) $\frac{Mg}{2k} + x_0$

(C) $\frac{2Mg}{k} - x_{\mathrm{D}}$

(D) $\frac{2Mg}{k} + x_0$

Q.18

The potential energy of a $1\ \mathrm{kg}$ particle free to move along the x -axis is given by $V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)\ \mathrm{J}$. The total mechanical energy of the particle is $2\ \mathrm{J}$. Then, the maximum speed (in $\mathrm{m/s}$) is

(A) 2

(B) $3/\sqrt{2}$

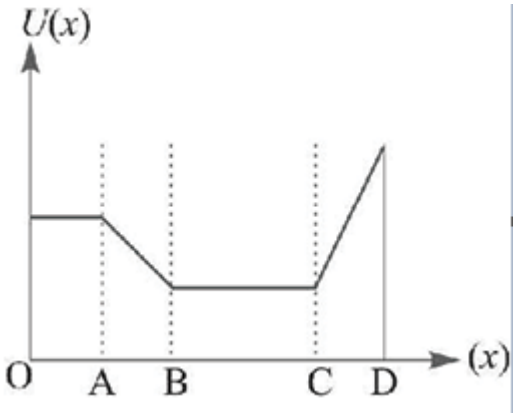
(C) $\sqrt{2}$

(D) $1/\sqrt{2}$

Q.19

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The figure gives the potential energy function $U(\mathrm{x})$ for a system in which a particle is in onedimensional motion. In which region the magnitude of the force on the particle is greatest:



(A) OA

(B) AB

(C) BC

(D) CD

Q.20

Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. What is the change in its kinetic energy (in joule) between $t = 2$ s and $t = 4$ s?

(A) 48

(B) 44

(C) 42

(D) 46

Answers & Solutions

Q.1 Answer:

$\mu g\sqrt{\frac{(2M+m)m}{kM}}$

Solution:

For the smaller block to move $kx_0 = \mu \mathrm{mg}$ **and from work energy theorem** $-\mu Mgx_0 - \frac{1}{2}kx_0^2 = -\frac{1}{2}Mv_0^2$

$+\mu Mg\left(\frac{\mu mg}{k}\right) + \frac{1}{2}k\left(\frac{\mu mg}{k}\right)^2 = \frac{1}{2}Mv^2$

$v = \mu g\sqrt{\frac{(2M+m)m}{kM}}$

Q.2 Answer:

$\frac{1-\mathrm{e}}{1+\mathrm{e}}$

Solution:

As $\mathrm{u}_2 = 0$ **and** $\mathrm{m}_1 = \mathrm{m}_2$**, therefore from** $\mathrm{m}_1\mathrm{u}_1 + \mathrm{m}_2\mathrm{u}_2 = \mathrm{m}_1\mathrm{v}_1 + \mathrm{m}_2\mathrm{v}_2$

we get $u_1 = v_1 + v_2$

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Also, $e = \frac{v_2 - v_1}{u_1} = \frac{v_2 - v_1}{v_2 + v_1} = \frac{1 - v_1/v_2}{1 + v_1/v_2}$,
which gives $\frac{v_1}{v_2} = \frac{1-e}{1+e}$

Q.3 Answer:

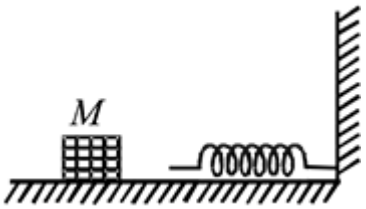
$\sqrt{MkL}$

Solution:

$\frac{1}{2}Mv^2 = \frac{1}{2}kL^2$

$\Rightarrow v = \sqrt{\frac{k}{M}} \cdot L$

$$\text{Momentum} = M \times v = M \times \sqrt{\frac{k}{M}} \cdot L$$
$$= \sqrt{kM} \cdot L$$



Q.4 Answer:

$1.2\ \text{kg}$

Solution:

Let $m = m1$ **and** $M = m2$

$m_1 = 2\ \text{kg}$

$u_1 = 4\ \text{m/s}$

$v_1 = \frac{u_1}{4} = \frac{4}{4} = 1\ \text{m/s}$

$u_2 = 0$

since nothing is given so we would take it as elastic collision. hence conservation of momentum is given by:

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$2 \times 4 + 0 = 2 \times 1 + m_2v_2$

$m_2v_2 = 6$

according to the conservation kinetic energy:

$\frac{1}{2}\,m_1u_1^2 + \frac{1}{2}\,m_2u_2^2 = \frac{1}{2}\,m_1v_1^2 + \frac{1}{2}\,m_2v_2^2$

$2 \times 16 + 0 = 2 \times 1 + m_2v_2^2$

$m_2v_2^2 = 30$

$\frac{m_2v_2^2}{m_2v_2} = \frac{30}{6}$

$v_2 = 5\ \text{m/s}$

$m_2 = 1.2\ \text{kg}$

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Q.5 Answer:

$200\ \mathrm{m/s}$

Solution:

Initial, K.E. $= \frac{1}{2}\mathrm{mv}^2 = \frac{1}{2} \times \frac{20}{1000} \times 600 \times 600$

$= 3600\ \mathrm{J}$

Change in K.E. = P.E. $\frac{1}{2}m\left(v^2 - v'^2\right) = mgh$

$\Rightarrow 3600 - \frac{1}{2} \times \frac{20}{1000} \times \mathrm{v}_1^2 = 4 \times 10 \times 80$

$\Rightarrow \mathrm{v}_1 = 200\ \mathrm{m/s}$

Q.6 Answer:



Solution:

$dU_{(x)} = -Fdx$

$\therefore \quad U_x = -\int_0^x Fdx = \frac{kx^2}{2} - \frac{ax^4}{4}$

$U = 0$ **at** $x = 0$ **and at** $x = \sqrt{\frac{2k}{a}}$; $\Rightarrow$ **we have potential energy zero twice (out of which one is at origin). Also, when we put** $x = 0$ **in the given function, we get** $F = 0$**. But** $F = -\frac{dU}{dx} \Rightarrow$ **At** $x = 0$; $\frac{dU}{dx} = 0$ **i.e. the slope of the graph should be zero. These characteristics are represented by (d).**

Q.7 Answer:

$0.67\ \mathrm{J}$

Solution:

Initial kinetic energy of the system

$K.E_i = \frac{1}{2}mu^2 + \frac{1}{2}M(0)^2 = \frac{1}{2} \times 0.5 \times 2 \times 2 + 0 = 1J$

For collision, applying conservation of linear momentum $\mathrm{m} \times \mathrm{u} = (\mathrm{m} + \mathrm{M}) \times \mathrm{v}$

$\therefore 0.5 \times 2 = (0.5 + 1) \times \mathrm{v} \Rightarrow \mathrm{v} = \frac{2}{3}\ \mathrm{m/s}$

Final kinetic energy of the system is $\mathrm{K} \cdot \mathrm{E_f} = \frac{1}{2}(\mathrm{m} + \mathrm{M})\mathrm{v}^2 = \frac{1}{2}(0.5 + 1) \times \frac{2}{3} \times \frac{2}{3} = \frac{1}{3}\ \mathrm{J}$

$\therefore$ **Energy loss during collision** $= \left(1 - \frac{1}{3}\right)\ \mathrm{J} = 0.67\ \mathrm{J}$

Q.8 Answer:

$\sqrt{2g(2R - h)}$

Solution:

For minimum v**, velocity of ball at the topmost point will be zero.**

By conservation of energy,

$\frac{1}{2}mv^2 = mg(2R - h)$

$v = \sqrt{2g(2R - h)}$

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Q.9 Answer:

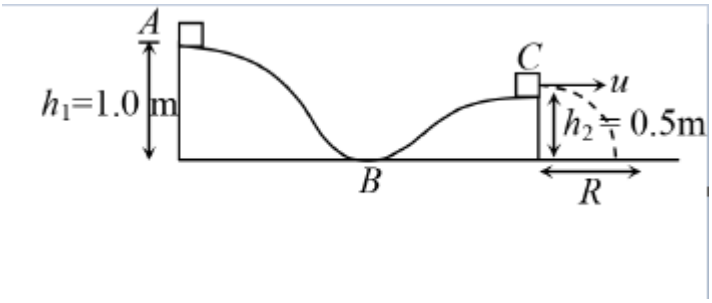
1.0 m

Solution:

$u = \sqrt{2g\,(h_1 - h_2)}$

$R = u\sqrt{\frac{2h_2}{g}} = 2\sqrt{h_2\,(h_1 - h_2)}$

$R = 2\sqrt{0.5 \times 0.5} = 1.0$ m

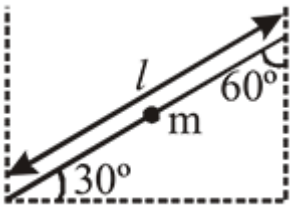


Q.10 Answer:

$\frac{mg\ell}{4}$

Solution:

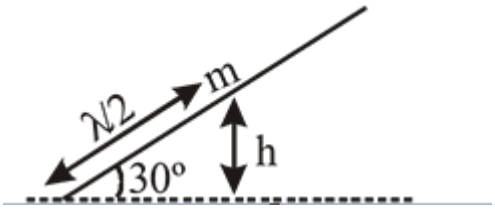
For any uniform rod, the mass is supposed to be concentrated at its centre.



$\therefore$ **height of themass from ground is,** $h = (l/2)\sin 30^\circ$

$\therefore$ **Potential energy of the rod** $= m \times g \times \frac{\ell}{2}\sin 30^\circ$

$= m \times g \times \frac{\ell}{2} \times \frac{1}{2} = \frac{mg\ell}{4}$



Q.11 Answer:

10.0 m

Solution:

$R_{max} = \frac{u^2}{g} at\theta = 45^0 = \left(\frac{mt^2}{2}\right)\left(\frac{2}{mg}\right) = \left(\frac{1}{2}kx^2\right)\left(\frac{2}{mg}\right)$

$\left(\frac{1}{2}mu^2 = \frac{1}{2}kx^2\right)$

Substituting the values $R_{man} = \frac{(600)\left(5\times10^{-2}\right)^2}{(15\times10^{-3})(10)} = 10.0$ m

Q.12 Answer:

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The maximum extension of the spring is $\frac{4mg}{k}$

Solution:

(a) At maximum extension $V_A = V_B = 0$

$2mgx = \frac{1}{2}kx^2 \quad x_n = \frac{4mg}{k}$

(b) $2mg\frac{2mg}{k} = \frac{1}{2}k\left(\frac{2mg}{k}\right)^2 + \frac{1}{2}(m+2m)v^2$

$v = 2g\sqrt{\frac{m}{3k}}$

(c) Net upward force $\Rightarrow \frac{4mg}{k} \times k = 4mg$

net downward force $\Rightarrow 2\text{mg}$

$a = \frac{4mg-2mg}{3m} = \frac{2}{3}g$

(d) for $x = \frac{2mg}{k}, a = 0$

$\therefore \quad T = 2mg$

Q.13 Answer:

$\frac{2m_1^2g^2}{k}$

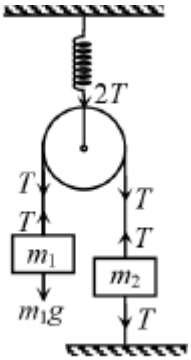
Solution:

$T = m_1g$

$kx = 2T = 2m_1g$

$x = \frac{2m_1g}{k}$

Energy stored $= \frac{1}{2}kx^2 = \frac{1}{2}k\frac{4m_1^2g^2}{k^2} = \frac{2m_1^2g^2}{k}$



Q.14 Answer:

$1 : 9$

Solution:

Let E **be the total energy then** $\frac{P.E}{K.E} = \frac{mgh}{E-mgh} = \frac{2}{3} \Rightarrow \text{E} = \frac{5}{2}\text{mgh}$

When velocity is double then inital energy becomes $4E$**.**

So, $\frac{mgh}{4E-mgh} = NL = \frac{mgh}{10mgh-mgh}$

On solving we get $\frac{P.E}{K.E} = \frac{1}{9}$**.**

Q.15 Answer:

1.9 J

Solution:

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$$J = m\left(v_f - v_i\right) \text{ or } 7.5 = 15\left(v_f - 0\right)$$
$$\Rightarrow v_f = 0.5 \text{ m/s}$$
$$\therefore \quad K = \frac{1}{2}mv^2 = \frac{1}{2} \times 15 \times 0.5^2 = 1.9 \text{ J}$$

Q.16 Answer:

$K_1 < K_2$

Solution:

Work done against friction force,

$W_f = \mu mg\cos\theta \times \frac{h}{\sin\theta} = \mu mgh\cot\theta$

As $\cot 30^\circ > \cot 60^\circ$

$\therefore K_1 < K_2$

Q.17 Answer:

$\frac{Mg}{2k} - x_0$

Solution:

If block M **moves a distance of** x **the extension in spring increases by** $2x$

By work energy theorem,

$$\frac{1}{2}k\left[x_0^2 - \left(x_0 + 2x\right)^2\right] + Mgx = O$$
$$-\frac{1}{2}k\left(4x^2 + 4xx_0\right) + Mgx = O$$
$$\frac{1}{2}k\left(4x + 4x_0\right) = Mg$$
$$x + x_0 = \frac{Mg}{2k}$$
$$x = \frac{Mg}{2k} - x_0$$

Q.18 Answer:

$3/\sqrt{2}$

Solution:

Maximam kinetic energy $= 2-$ **(minimum potential energy)** $\frac{1}{2}mv^2 = 2 - \left(-\frac{1}{4}\right) \quad \left(\because U_{\min} = -\frac{1}{4}J\right.$ **at** $\left.x = \pm 1\right)$

$\frac{1}{2}mv^2 = \frac{9}{4} \Rightarrow v = \frac{3}{\sqrt{2}}$ m/s

Q.19 Answer:

CD

Solution:

$|F| = \frac{dU}{dx}$**, which is greatest in the reagion** CD**.**

Q.20 Answer:

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Solution:

$$P = 3t^2 - 2t + 1 = \frac{dE}{dt}$$

$$\therefore dE = (3t^2 - 2t + 1)dt$$

$$E = \int_{t=2s}^{t=4s} (3t^2 - 2t + 1)dt$$

$$= \frac{3t^3}{3} - \frac{2t^2}{2} + t \Big|_{t=2s}^{t=4s}$$

$$= [4^3 - 2^3 - 4^2 - 2^2 + (4 - 2)]$$

$$E = 56 - 12 + 2 = 46\ \text{J}$$

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